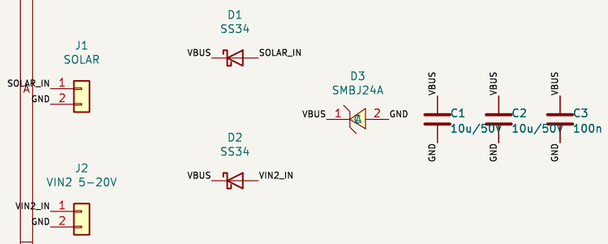
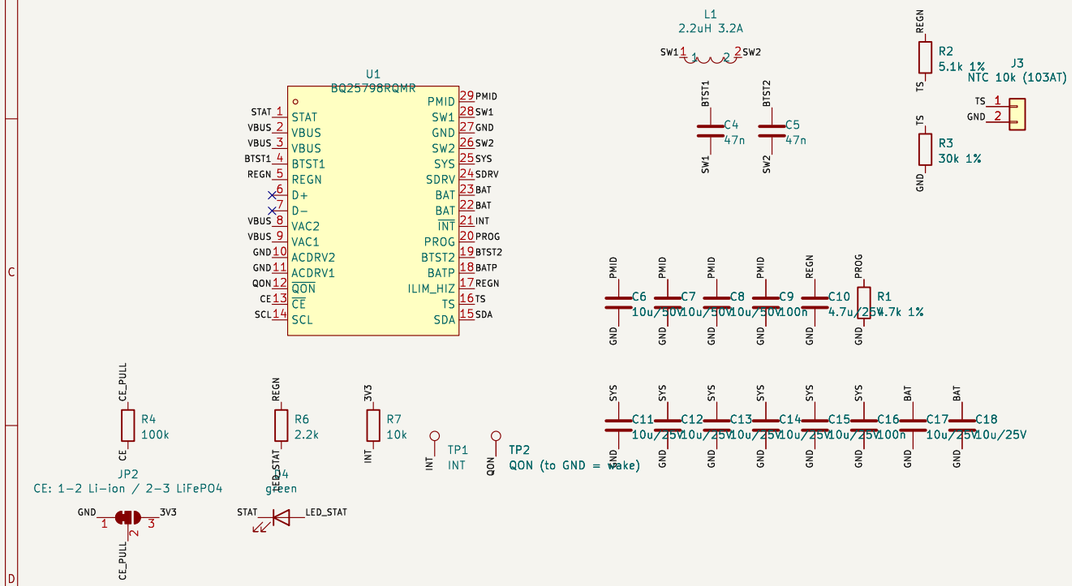
1. ВХОДОВЕ: соларен панел 5–22 V (Voc до 25 V на студ) и втори вход 5–20 V



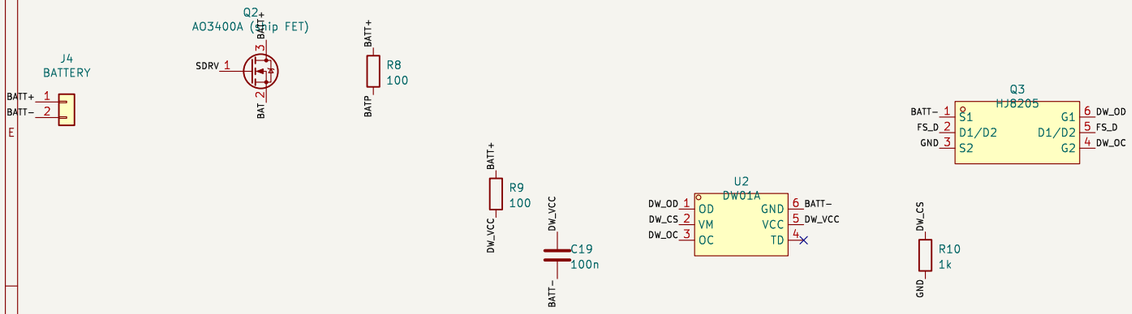
SS34 = OR на двата входа + защита от обратна полярност. VAC1=VAC2=VBUS, ACDRV към GND (без ACFET).

2. MPPT ЗАРЯДНО BQ25798 (1S, 750 kHz, 2.2 uH), I2C 0x6B



CE: JP2 1–2 (Li-ion) зарежда и без Xiao; 2–3 (LiFePO4) само след запис на VREG и DO=0. Watchdog off. D+/D– не се ползват.

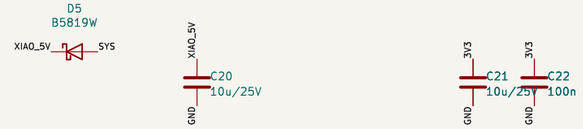
3. БАТЕРИЯ 1S (Li-ion по подразбиране, LiFePO4 с настройка): ship FET + DW01A/HJ8205



Q2: D към батерията, S към BAT: в shutdown (SDRV_CTRL=01, 0.5 uA) диодът на Q2 спира разреждането, зарежда се само когато се появи напрежение от панела. DW01A: 2.4 V / 4.3 V резерв, HJ8205 в минуса.

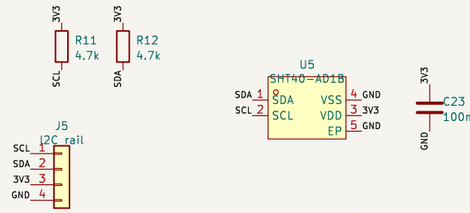
4. XIAO nRF52840 + Wio-SX1262 (пинове като variants/xiao_nrf52)



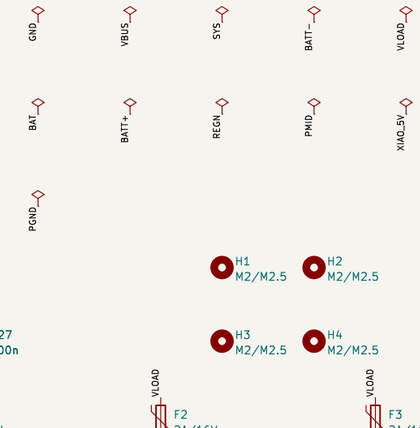


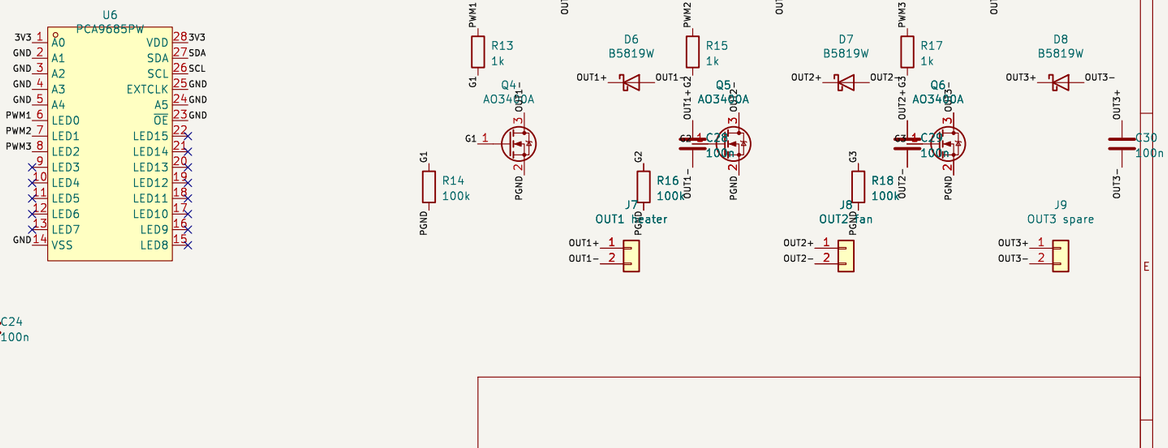
D5: SYS към пина 5V на Xiao (USB на Xiao не захранва SYS). 3V3 идва от регулатора на Xiao.

5. I2C (SCL=D6, SDA=D7): BQ25798 0x6B, SHT40 0x44, PCA9685 0x41



6. ИЗХОДИ ЗА ТОВАРИ (PWM, до 2 A): нагревател, вентилатор, резерв





n3tk0/MeshCore		
Sheet: /		
File: xiao-solar.kicad_sch		
Title: Xiao nRF52840 + Wio-SX1262 solar board		
Size: A3	Date: 2026-10-29	**Rev: 0.1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1